Problem Statement	Logistics teams lack real-time visibility into shipment status, delays, and exceptions, leading to constant follow-ups and poor decision-making.
Key Goal:	To design a tracking dashboard that displays **live shipment status**, provides **alerts**, and delivers **actionable insights.**

Understanding Yatra Freight

After capturing significant market share in the corporate travel business, Yatra launched Yatra Freight in October 2020. The platform handles ocean and air freight shipments, as well as customs clearance and documentation. Yatra Freight serves manufacturers and exporters who need reliable international shipping without the usual opacity of traditional freight forwarders.

Business Model

Yatra Freight has adopted an aggregator business model, connecting shippers directly to ocean carriers and airlines (B2B).

Revenue Sources

Freight Booking Margins + Custom Clearance Fees + Documentation & Processing Fees + Platform Service Fees

User Base

- **Import dependent Businesses**
- **Export Oriented Businesses**
- **E-Commerce & D2C Brands**
- **Corporate Travel Clients**
- **SME Manufacturers**
- **Distributors & Trading Houses**

What is the true problem?

Yatra Freight lacks real-time shipment visibility, forcing teams to manually chase status updates via phone calls, emails, and fragmented carrier portals.

Business Impact:

- High Operational cost in manual tracking & coordination.
- Customer churn risk due to lack of visibility & delayed communication
- Creates a scaling bottleneck; onboarding more customers will require proportional employee growth

Key Users:

- **Operations Team**
- **Sales Team**
- **Customer Support Team**

Why should we solve this now?

- Manual tracking creates a scaling bottleneck; shipment volume will grow faster than Ops/Support capacity if we don't automate visibility.
- A dashboard shifts teams from "status chasing" to "exception handling," enabling non-linear growth.
- Acting now creates a clear edge over traditional forwarders; delaying gives competitors time to match visibility and erodes differentiation.

Business Objective:

To build an internal tracking dashboard that provides live shipment status, detects any delays/exceptions early, minimizes manual interventions and decision latency across the operations team, customer support team, as well as sales team, and also delivers actionable insights (like delay patterns, recurring delay/exception reason, etc), to improve future decision making, improving operations processes to increase efficiency.

In Scope Requirements:

- To enable **end-to-end visibility of the current state of each shipment** within a unified view of the shipment lifecycle, with all internal teams having access to the same unified view of the state of each shipment.
- To enable **monitoring of critical milestones** (ETD, ETA, and ATA) as well as any deviations in the process.
- Enable **early detection and prioritization of delays/exceptions** so teams can act ahead of escalations.
- Provide role-aligned **access to shipment data** so that Operations, Customer Support, and Sales teams can perform their functions with minimal cross-functional dependencies.
- Enable **reduction in manual coordination** for follow-ups, thereby decreasing operational costs.
- To facilitate accountability and traceability through a chain of key actions and communications associated with shipment status changes and exceptions.
- Enable **actionable operational insights** (e.g., patterns by route/lane, stage, exception type, carrier/vendor) & use them to improve future operations continuously.
- Enable **automated shipment-stage notifications,** so key internal stakeholders (Operations and Customer Support) receive timely email updates when a shipment changes stage or becomes at risk/delayed, ensuring consistent communication and faster response to exceptions.

Out of Scope Requirements:

- Not to focus on providing a **customer-facing portal or customer login experience for shipment tracking** & **documents access** (customer self-service is a separate product area).
- The business will not take **ownership of or optimize the physical execution** (warehouse/yard operations, last-mile delivery) for this problem.
- Do not aim to **guarantee perfect real-time location accuracy** across all legs as a business promise

Value generated for Business by solving this problem?

- Drastic reduction in manual tracking costs and coordination overhead.
- Reduces Churn rate & boosts retention & NPS through transparent, real-time communication
- Establishes a competitive advantage over traditional, non-digital freight operators
- Enables the Ops team to manage higher shipment volumes without proportionate employee growth.
- Unlocks actionable insights from data on dashboard on carrier performance and route optimization.

Functional Requirements:

Shipment Display:

- Dashboard shall display a paginated list of all active shipments.
- Each listed shipment must display: Shipment ID, Customer Name, Route (Origin/Dest), Current Stage (e.g., In Transit), Status Flag (On Track / At Risk / Delayed), and Last Updated Timestamp.
- The dashboard should support sorting by ETA (soonest first), Status (Delayed first), and Last Updated (oldest/newest).
- The Dashboard should support "Quick Filters" for: Import/Export, Status (On Track/Delayed), Mode (Air/Ocean).
- The dashboard should provide a global search bar that accepts Shipment ID, AWB/BL Number, PO Number, or Customer Name.
- Dashboard should display a visual "Progress Bar" showing the unified stages (Booked > Picked Up > ... → Delivered) with the current stage highlighted.
- Dashboard shall display a "Milestone Table" showing Planned vs. Actual dates for ETD, ETA, and ATA. & highlight any milestone deviation (e.g., if Updated ETA > Planned ETA) in red.
- Dashboard should display a map widget showing the shipment's current (or last known) location based on status/coordinates

Map View (Global Visibility):

- Dashboard should have a "Global Map View" plotting all active shipments as pins.
- Pins shall be color-coded by status (Green=On Track, Red=Delayed, Yellow=At Risk).
- Clicking a pin shall open a "Quick View" card with Shipment ID, Route, and Status, with a link to the full Detail View.

Exception & Delay Management:

- System should automatically flag a shipment as "Delayed" if the current date exceeds the Planned ETA or a critical milestone is missed & notify Ops team with an email
- The system shall allow Operations users to manually create an "Exception Alert" for a shipment. When creating an exception, the system requires the following: Exception Type (e.g., Customs Hold, Carrier Delay), Severity (Low/Medium/High), and Internal Note
- The system shall display active exceptions prominently on the Shipment Detail view until they are resolved & allow Ops to "Resolve" a exception, capturing a note and timestamp.

Emails & Communications:

- Dashboard should allow Support/Sales/Ops users to "Send Update" from the Shipment Detail view. "Send Update" feature shall generate a templated email including: Current Status, Latest ETA, and (optional) Exception Reason.
- System shall automatically trigger an internal email notification to the assigned Ops owner when a High-Severity exception is raised.
- Dashboard should maintain a "Communication Log" on the shipment record, displaying all system-sent and user-sent emails (Timestamp, Sender, Recipient).

Role-Based Access Control:

- **Operations Users:** Can View/Edit all shipments, Create/Resolve Exceptions, and Override Milestones.
- Customer Support Users: Can view all shipments, Send Update Emails. Read-only access to milestones/exceptions.
- **Sales Users:** Can view shipments (optionally restricted to their assigned accounts), Send Update Emails. Read-only access.

NFRs:

Search results must load in < 2 sec.

data must reflect the latest source update within 15 minutes

Dashboard Landing Page:

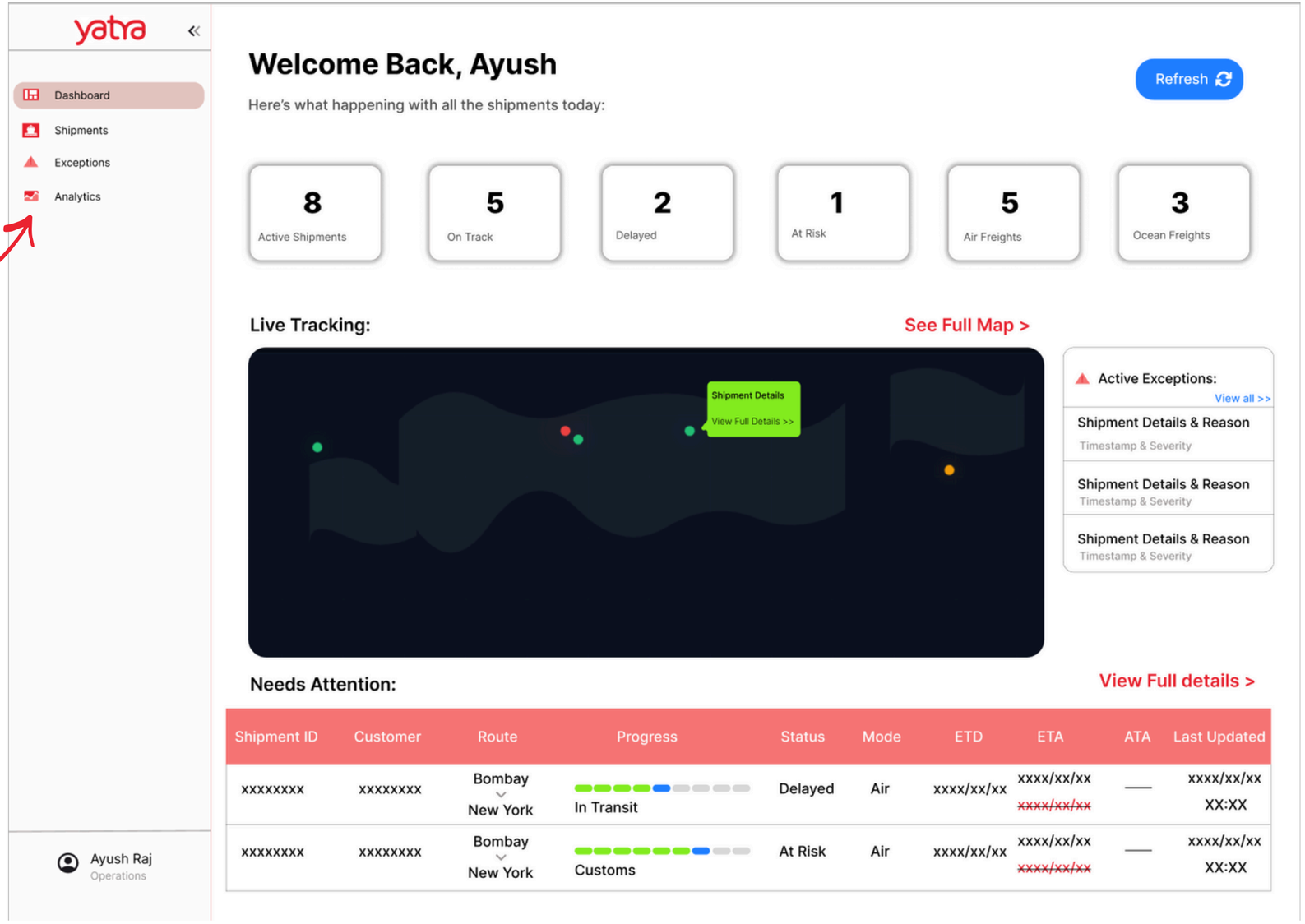
The dashboard landing page provides a live snapshot of operational health. Displays real-time KPIs (Active Shipments, On Track, Delayed, etc.), a global map view with color-coded shipment pins (green=on track, red=delayed), and a "Needs Attention" table highlighting at-risk/delayed shipments. Users can quickly identify problem areas and drill into details without scrolling through hundreds of shipments.

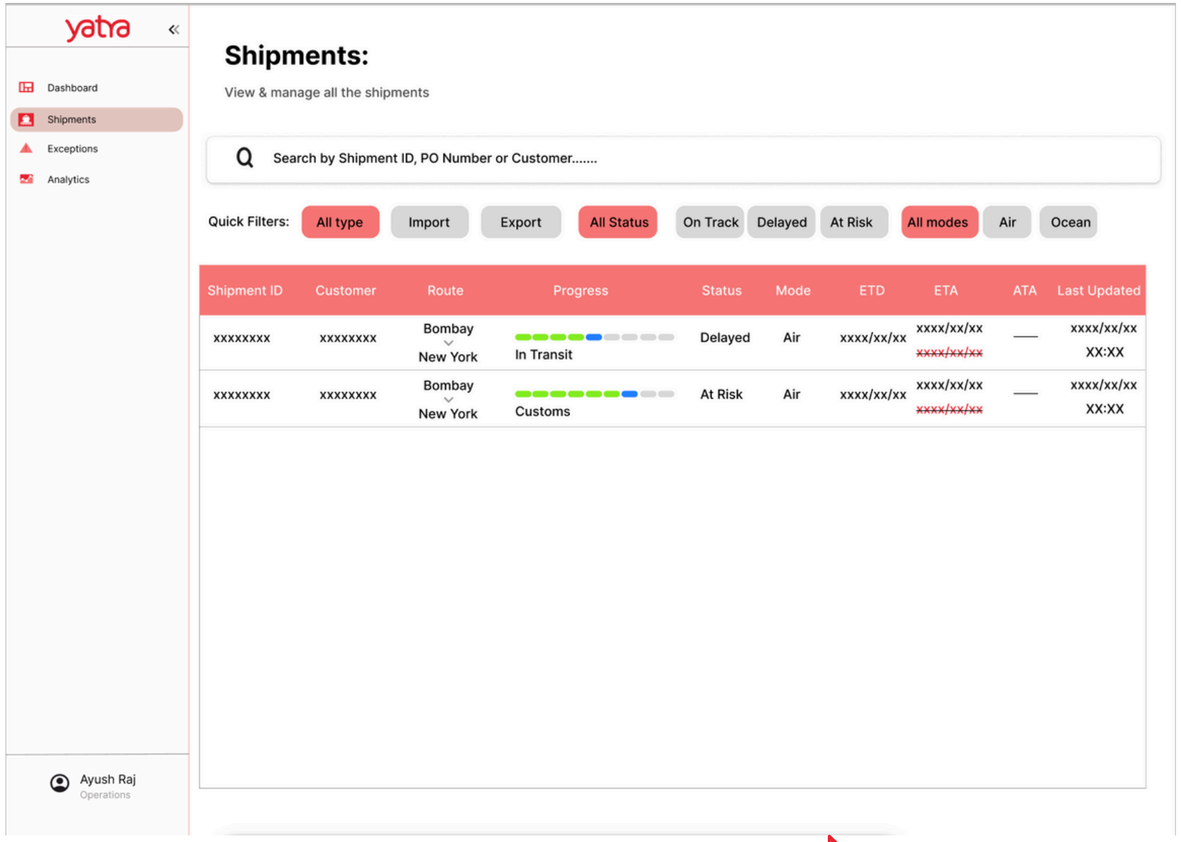
One-click access to historical performance reports: view carrier reliability trends, top delay reasons by lane, and other analytics that provide actionable insights."

Key Workflows:

- **OPS Team:** Initiate & Setup > Tracks Origin Execution > Manage Exception & Delay > Transit & Milestone Updates > Destination & Delivery Coordination > Close shipment record
- **Sales Team:** Filter by Key accounts > Monitor high priority shipments > Track Milestones > Proactively inform to manage expectations > Review Performance
- **Customer Support Team:** Fast Search & Identify > read latest status/ETA + reason > send update email > Solves query > escaletes complex query (if needed)

Link for Wireframes



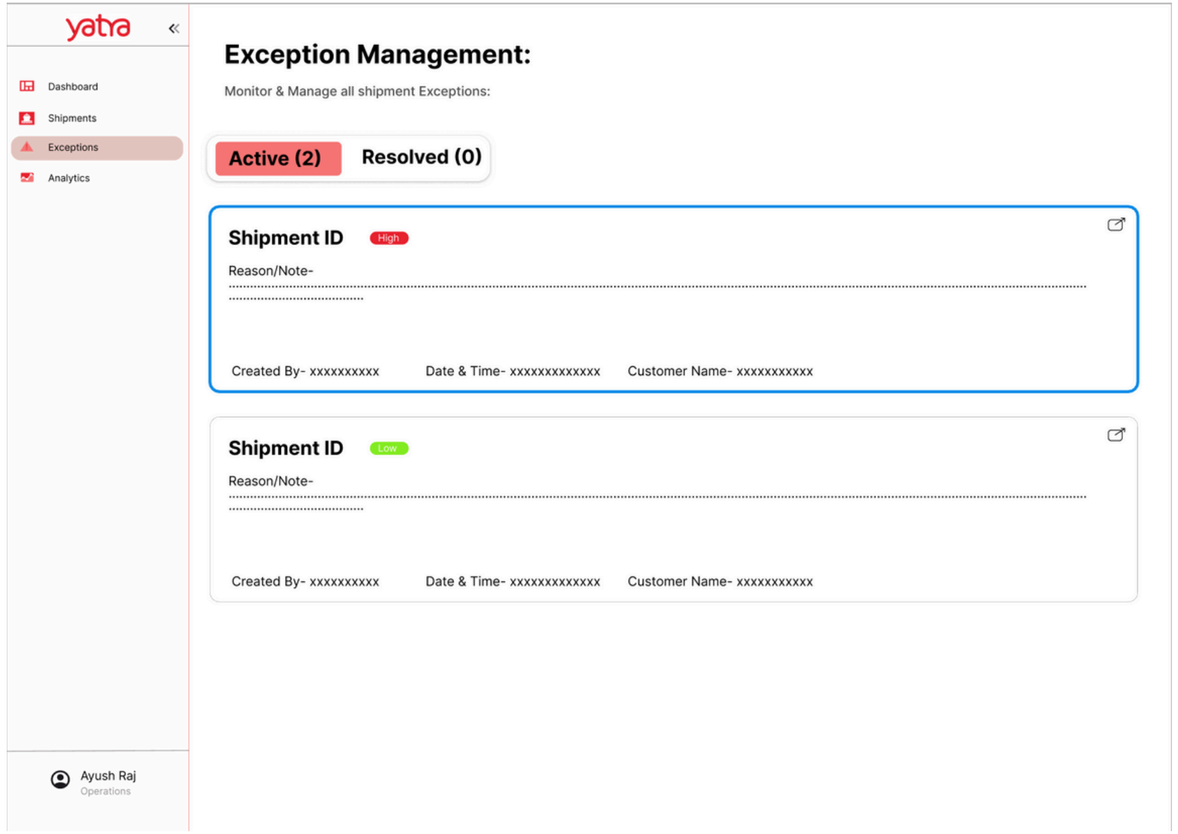


Shipments Details Page:

The core interface where Ops, Support, and Sales manage workload. Displays a searchable, filterable list of all shipments with key columns: Shipment ID, Customer, Route, Progress (visual bar), Current Stage, Status flag (On Track/Delayed/At Risk), Mode (Air/Ocean), Milestones (ETD/ETA/ATA), and Last Updated. Quick-access filters (All types, Import/Export, Status) allow teams to segment by priority without leaving the list.

Exceptions page:

The exception workflow hub where Ops monitors, assigns, and resolves issues. Shows a tabbed view of Active and Resolved exceptions. Each exception displays the Shipment ID, Reason Note (a text field for context), Created By, Date & Time, and Customer Name. Ops can quickly review exception details, add action notes, and mark resolved when the issue is fixed. Maintains audit trail for accountability and enables quick escalation to relevant teams.



Metric Name	Calculation Method	Why it Matters?
Proactive exception handling rate (NSM)	(# exceptions where Ops resolved **or** Support sent update **before** customer escalation) / (Total exceptions)	The core value prop is shifting from reactive "chasing" to proactive management. This drives trust and retention.
DAU % internal (Guardrail)	Unique users who performed at least 1 meaningful action (search, update, filter) per day / Total eligible employees	Validates that dashboard is replacing manual workflows. If adoption is low, the tool isn't useful or trustworthy.
Average Resolution Time for "At Risk" Alerts	Avg(Exception resolved time – Exception created time)	Directly measures if the Dashboard view helps Ops clear bottlenecks faster. Lower time = faster corrective action.
"Where is my Shipment" Ticket Volume	Total # of support tickets tagged "Status Inquiry" / Total Active Shipments (Compare pre-launch vs. post-launch trend)	If the internal team is proactive (using the dashboard notifications), inbound customer queries should drop.
Data freshness compliance	Shipments with last update within X hours (by stage/mode) / Total shipments	Prevents teams from reverting to calls/emails; fresh data is key to being a “single source of truth.”
Milestone completeness (ETD/ETA/ATA coverage)	Shipments with ETD, ETA, ATA present / Total active shipments	If milestone data isn’t available, neither alerts nor customer updates can be reliable; milestones are explicitly required.
Alert quality (Guardrail)	(Alerts marked “Not actionable/False alarm” or closed with reason “no issue”) / (Total alerts created)	Prevents alert fatigue; if noise is high, teams will ignore alerts and revert to manual tracking.

Assumptions:

- The dashboard will be used exclusively as an **internal tool by Yatra Freight's Operations, Customer Support, and Sales teams** (not exposed to external customers or partners). This allows us to focus on **internal workflows, exception management, and team communication** without managing external authentication, data privacy/compliance, or customer-facing SLAs.
- Carriers (ocean lines, airlines) and logistics partners (CHAs, freight forwarders) will **provide shipment updates (stage changes, milestones)** either through API integrations or periodic feeds with reasonable frequency
- **Historical data (past shipments, customer records, routes) will be available** in Yatra Freight's backend systems and can be synced into the dashboard database without major migration friction.
- Operations, Customer Support, and Sales teams will actively **use the dashboard as their primary source of truth** for shipment status (not continue reverting to phone calls/emails if milestones are stale).

Trade-Offs:

#1 Internal Portal vs Customer facing portal

- We're not building a self-serve customer tracking portal like FedEx/DHL offers (dilutes focus, adds an API stability burden).
- Instead, Ops/Support uses the dashboard to communicate proactively, indirectly improving customer experience.

#2 Granular Sub-stages vs. Simplicity

- We're not tracking every micro-stage (gate-in, weigh, load) because carriers don't provide that data.
- Instead: Unified, mode-agnostic stage model (10 key stages: Booked → Delivered) that's easy to understand and automate.

#3 Manual Overrides vs. Data Purity

- We allow Operations to override milestones (Ops need to correct carrier errors quickly).
- But: All overrides are audit-logged for accountability and transparency.

#4 Map Visualization vs. Accuracy

- The map shows color-coded shipment clusters, not pinpoint coordinates (many shipments are "at port/airport," not exact GPS points).
- Instead: Focus on geographic triage for Ops (identify regional bottlenecks), not navigation.

Product Assignment

Problem Statement 1 Ayush Raj BITS Goa